

LoRA

$$\begin{array}{c}
 B \\
 \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \\
 d_{\text{out}} \times r
 \end{array}
 \times
 \begin{array}{c}
 A \\
 \begin{array}{|c|c|c|c|c|} \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \end{array} \\
 r \times d_{\text{in}}
 \end{array}
 =
 \begin{array}{c}
 \Delta W_{\text{LoRA}} \\
 \begin{array}{|c|c|c|c|c|} \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \end{array} \\
 \text{rank}(\Delta W_{\text{LoRA}}) \leq r
 \end{array}$$

TiRA

k : group index ($0 \leq k < K$); m : block-row index ($0 \leq m < M$)

$$\begin{array}{c}
 b_{k,m} \\
 \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \\
 n_{\text{out}} = d_{\text{out}}/M
 \end{array}
 \times
 \begin{array}{c}
 a_{k,m}^{\top} \\
 \begin{array}{|c|c|c|c|c|} \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \end{array} \\
 n_{\text{in}} = d_{\text{in}}/M
 \end{array}
 =
 \begin{array}{c}
 C_{k,m} \\
 \begin{array}{|c|c|c|c|c|} \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \square & \square & \square & \square & \square \\ \hline \end{array}
 \end{array}$$

place $C_{k,m}$ at block $(m, (m+k) \bmod M)$

$$\begin{array}{c}
 \Delta W_k \\
 \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \\
 \text{rank}(\Delta W_k) = M
 \end{array}$$

$$\begin{array}{c}
 \Delta W_0 \\
 \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \\
 \Delta W_1 \\
 \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \\
 \Delta W_2 \\
 \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array}
 \end{array}
 + \dots \Rightarrow
 \begin{array}{c}
 \Delta W_{\text{TiRA}} \\
 \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \square & \square & \square & \square \\ \hline \end{array} \\
 \text{rank}(\Delta W_{\text{TiRA}}) \leq KM
 \end{array}$$